

Performance Characterization of a Low Reynolds Number Micro-Nozzle Flow

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With the rise of CubeSat usage there is a growing use-case for low Reynolds number nozzles as a component within micro-propulsion systems. However, design of such systems is difficult due to the viscous losses encountered in such low Reynolds number flow regimes. As part of a program to develop a design tool for efficient micro-nozzle design, the performance of a test nozzle is measured across a range of Reynolds numbers to create an experimental database for model validation. The thrust and specific impulse efficiency of are measured using a torsional thrust balance at various flow rates to cover the Reynolds number range of 60 – 500 for 300 K nitrogen.

Nomenclature

A_{eff}	= effective area of the capacitor plate
c_d	= damping coefficient
C	= capacitance of the calibration force plates
d_0	= initial gap between calibration force plates
d^*	= nozzle throat diameter
f	= vibration frequency
f_n	= system natural frequency
F_{CFG}	= force generated by the capacitance force generation plates
F_T	= thrust produced by the nozzle
I_{sp}	= specific impulse
k	= linear stiffness
k_θ	= torsional stiffness of thrust stand
K_T	= thrust stand calibration constant
m	= system mass
$\dot{m}$	= propellant mass flow rate
N	= number of measurements taken
p_0	= thrust chamber pressure
p_e	= nozzle exhaust pressure
r_a	= distance of the actuator from the thrust stand axis of rotation
r_e	= distance of the force calibration plates from the thrust stand axis of rotation
r_o	= distance of the optical sensor from the thrust stand axis of rotation
r_T	= distance of the nozzle test bed from the thrust stand axis of rotation
R	= gas specific constant
Re^*	= nozzle throat Reynolds number
T	= transmissibility of a system
T_0	= stagnation temperature in thrust chamber
V	= voltage across the calibration force plates
x_a	= position of actuator
x_o	= position as measured by the optical sensor
Z_s	= capacitance calibration zero-shift
γ	= ratio of gas specific heats
ϵ_0	= permittivity of free space
ζ	= damping ratio
η	= nozzle efficiency
μ	= propellant viscosity
τ	= torque on thrust stand arm

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I. Introduction

There is growing interest in the use of CubeSats to increase space accessibility for a variety of organizations that otherwise would be inhibited by the cost barriers associated with large vehicle launches. Additionally, CubeSats enable low-cost multi-vehicle mission concepts, such as orbital constellations for communications networks. However, current CubeSats generally contain little to no propulsive capability, which drastically limits orbital life. As an example, the Naval Research Laboratory (NRL) launched in 2010 the QbX CubeSat, and with no propulsion system the orbit had decayed within a month. Internal studies have found that the use of a small resistojets propulsion system could have extended this life-span to at least four months. This substantial increase in CubeSat utility is limited by the relatively small selection of flight-tested micro-propulsion options. Therefore, the development of low-power micro-propulsion systems is vital to the continued expansion of CubeSat utility.

While there are multiple design concepts for micro-propulsion, from FEEP to arcjet thrusters, one particularly appealing option for CubeSats is the resistojets. Two key design limitations when designing a CubeSat propulsion system are the limited power and space available. This generally excludes high-power systems or those that require advanced electronics (high voltage or microwave systems). The simplicity and scalability of micro-resistojets (MRJ) are thus advantageous for a CubeSat mission designer. One of the difficulties with designing resistojets systems for micro-propulsion application is for the thrust levels desired, the nozzle dimensions and flow rates are very small. This results in a very low Reynolds number flow characterized by highly viscous behavior. Whereas in large nozzles the boundary layer in the diverging section of the nozzle is negligible, for micro-propulsion nozzles the boundary layer can occupy a significant portion of the nozzle. As part of a program to develop a design tool that can optimize nozzle geometry for a given throat Reynolds number, high fidelity measurements of the resistojets performance are required to validate the modelling tools.

The first phase of model validation is to use the macroscopic performance of the nozzle and compare it to aggregate model results. For that to occur, the performance of multiple nozzle geometries must be measured across a range of operating conditions. There are multiple nozzle performance metrics available, but in the interests of accuracy only those parameters that can be directly measured or easily calculated with minimal uncertainty are used. For this work, the chosen performance metrics are thrust and specific impulse efficiency, which is defined as the ratio of the measured specific impulse of the nozzle flow to the expected specific impulse assuming isentropic expansion. In order to sufficiently cover the parameter space for model validation, multiple nozzle geometries are tested across a range of temperatures and pressures to cover the Reynolds number range of 100-2000. Previous work in this area has either been limited by relatively large uncertainties compared to nozzle performance variation,^{1,2} or did not have a sufficiently wide parameter space to cover the possible operating conditions.³⁻⁵ This work presents the experimental characterization of the thrust, specific impulse, thrust chamber pressure and temperature, and nozzle specific impulse efficiency for a conical nozzle across the Reynolds number range of 60-500 for nitrogen. This work is a portion of a larger program to create thruster nozzles design tools that perform numerical optimization of the nozzle geometry to minimize viscous losses in the nozzle for challenging viscous flow regimes; details of these efforts have been published by co-investigators Osborn,⁶ Holman,⁷ and Rosenberg.⁸ Future work will report results for other species and configurations.

II. Experimental Apparatus

A. Facilities

All testing is performed in NRL's South Vacuum Chamber, which is 2 m diameter by 2.3 m tall. The chamber is pumped with a 48" NRC diffusion pump with a nominal pumping speed of 100,000 l/s for air. The diffusion pump is backed by four Stokes 412H-10 piston vacuum pumps, each with a pumping speed of 300 cfm. Base pressure at high vacuum operation is 1.5×10^{-5} Torr. Pressure in the chamber is measured using a Bayard-Alpert type ion gauge in high vacuum, and rough vacuum pressure is measured using a Granville Phillips 275 convectron gauge.

B. Nozzle Test Apparatus

The nozzle flow being tested is generated in a modular nozzle test bed. Inspired from micro-resistojets design, the nozzle test bed consists of a stainless steel casing with welded Swagelok attachment points for a pressure transducer, a metal sheath-encased thermocouple, a cartridge heater, and a gas feed line. Gas is fed at the back of the device and exits through the nozzle at the opposite end of the test bed. An annular heat exchanger facilitates the heat transfer between the heater cartridge and the gas. The nozzle itself is machined into a nut that threads onto the end of the test bed. Gas temperature is measured using a K-type thermocouple with an uncertainty of $\pm 0.75\%$. Pressure inside the test bed is measured with a Taber 2403SAT pressure transducer that is calibrated in-house with a GE Model 7350 pneumatic pressure controller; the resulting uncertainty of the transducer after calibration is $\pm 0.0035\%$. Gas flow rate into the nozzle is measured using an Alicat MS-100 for flows below 125 sccm, and an Alicat MS-1000 for flows higher than 125 sccm, both with an uncertainty of 0.4% of the reading plus 0.2% of full scale. Figure 1 shows a picture of the nozzle test bed disassembled.

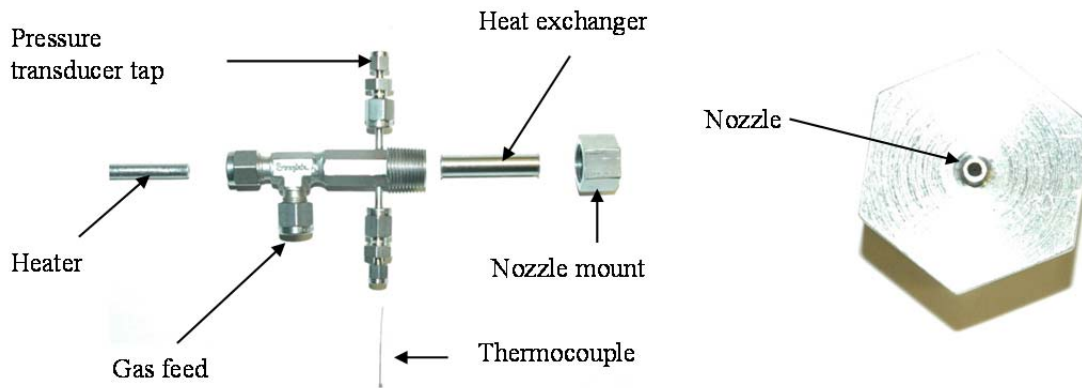


Figure 1. Nozzle test bed (left) and nozzle machined into nozzle mount (right).

The nozzle used for preliminary testing was designed to test the feasibility of using conventional machining (mill and lathe) for micro-nozzle fabrication. The nozzle is a conical de Laval nozzle with a 90° total angle in the converging portion, a 20° half-angle in the diverging region, a throat diameter of 0.006 in, and an area ratio of 100. However, complications in the fabrication resulted in a throat diameter of 0.0114 in, which gave an area ratio of 26. This demonstrated that at these small scales, in order to acquire the desired nozzle geometry with a high degree of fidelity, EDM processes such as die-sink and wire EDM would be required. However, for preliminary testing of the nozzle test-bed, the faulty nozzle was still sufficient. Inspection of nozzle throat and exit diameters was performed using a Zeiss Axio Zoom V16 digital microscope. Figure 2 shows a cross-sectional view of the nozzle geometry.

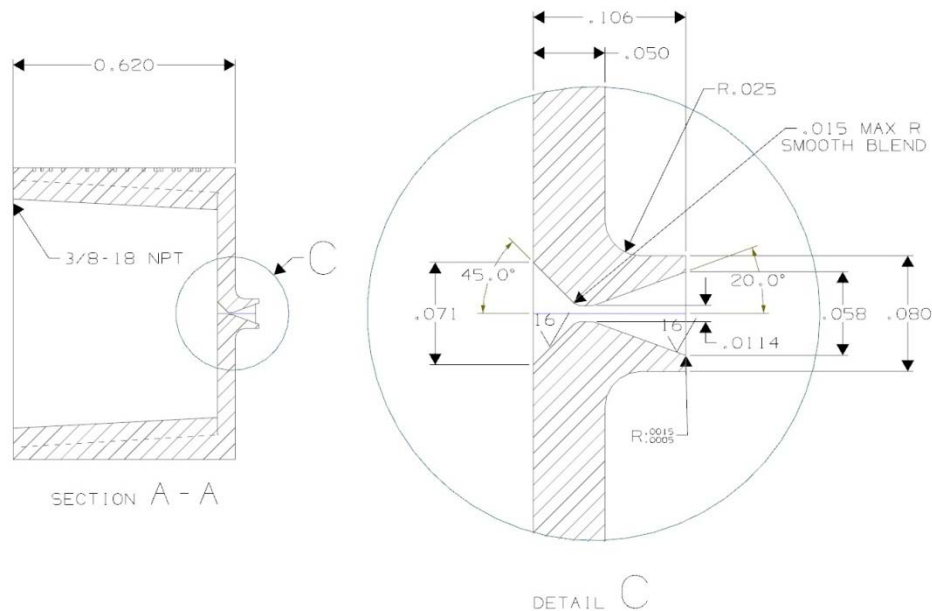


Figure 2. CAD model of conical nozzle.

C. Diagnostics

The thrust generated by the nozzle is measured using a torsional thrust balance developed in collaboration with Busek Co., Inc. The thrust stand has a nominal range of 0.1-10 mN with an uncertainty of $\pm 50 \mu\text{N}$. In situ calibration of the thrust stand is performed using capacitive force generation (CFG) plates across a range of voltages and comparing the calculated capacitive force to the torsional displacement as measured by a Philtec laser displacement sensor. The capacitive force is estimated based on measurements of the CFG plate capacitance using a BK Precision 889B LCR meter. Leveling of the thrust stand is verified using two piezoelectric tilt sensors arranged orthogonally along one of the base beams of the torsional stand. Two linear actuators connected to two corners of the thrust stand base provide fine control of the alignment.

C.1 Thrust Stand Calibration

Calibration of the thrust stand is performed in-situ using a pair of CFG plates. During the final stages of the pump-down, the capacitance of the CFG plates is measured as a function of distance between the two plates. The position of the

thrust stand arm is controlled with a linear actuator that is extended to be in direct contact with the arm. The capacitance data is curve fit to the equation below developed in conjunction with Busek,

$$C = \frac{\epsilon_0 A_{eff}}{d_0 - \frac{r_e}{r_a} \Delta x_a + Z_s} \quad (1)$$

where C is the capacitance, d_0 is the initial gap between the CFG plates, r_e and r_a are the radial distances from the arm axis of rotation for the CFG electrode plates and the actuator, respectively, Δx_a is the change in position of the actuator moving the arm, ϵ_0 is the permittivity of free space, A_{eff} is the effective area of the capacitor formed by the two plates, and Z_s is the zero-shift of the curve fit. The last two parameters are variables of the curve fit; the effective area approximates the area of a parallel plate capacitor if both plates had equal areas, and the zero-shift compensates for the fact that as the arm rotates, the plates are no longer strictly parallel.

Following the capacitance measurement, the voltage drop across the two CFG plates is swept from 0-800 V and the displacement of the arm is measured by the optical displacement sensor. The force generated by the CFG plates can be calculated using the equation

$$F_{CFG} = \frac{1}{2} \frac{\epsilon_0 A_{eff} V^2}{\left(d_0 - \frac{r_e}{r_o} \Delta x_o + Z_s \right)^2} \quad (2)$$

where V is the voltage across the CFG plates, r_o is the radial distance of the optical sensor from the axis of rotation, and Δx_o is the change in position as measured by the optical displacement sensor. Plotting the resulting CFG forces yields a linear function of Δx_o as long as thrust stand deflections are small, as shown below in Figure 3.

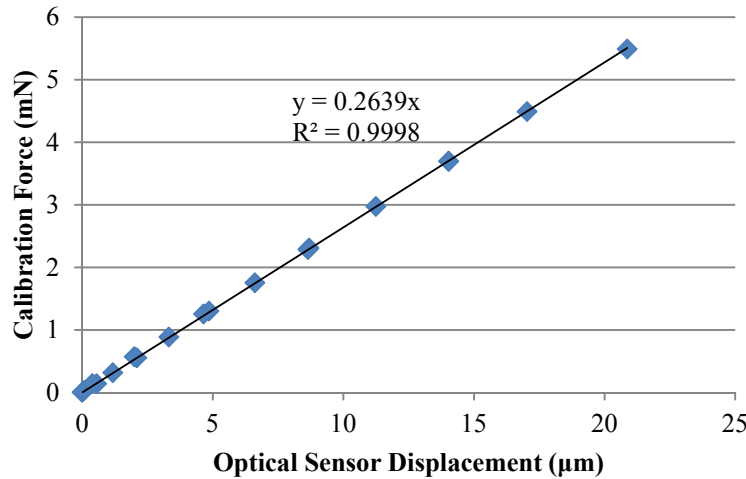


Figure 3. Calculated calibration force vs measured arm displacement.

The thrust stand torsional stiffness can be defined as the ratio of the torque, τ , on the arm to the angular displacement, $\Delta\theta$.

$$k_\theta = \frac{\tau}{\Delta\theta} = \frac{r_e r_o F_{CFG}}{\Delta x_o}$$

The value of the torsional stiffness can be calculated from the regression line in Figure 3. The torsional stiffness can in turn be used to define a calibration constant for the thrust stand, K_T , which gives the thrust in terms of the linear displacement of the thrust stand arm from the optical sensor.

$$F_T = \frac{k_\theta}{r_T r_o} \Delta x_o = K_T \Delta x_o \quad (3)$$

where T is thrust, and r_T is the radial distance of the thruster centerline axis to the axis of rotation of the arm.

C.2 Uncertainty Analysis

The uncertainty of the thrust measurement is the combination of the uncertainties of the optical position measurements, $U_{\Delta x}$, the calibration process, U_{cal} , and the gas flow, U_Q .

$$U_T = \sqrt{U_{\Delta x}^2 + U_{cal}^2 + U_Q^2} \quad (4)$$

The uncertainty of the position measurement is treated as the uncertainty of the differences between two average position measurements: with gas flow and without. This calculation is a function of the standard deviation of the position measurements, σ , and the number of measurements taken at that position, N .

$$U_{\Delta x} = \sqrt{\left(\frac{\sigma_{x1}}{\sqrt{N_1}}\right)^2 + \left(\frac{\sigma_{x0}}{\sqrt{N_0}}\right)^2} \quad (5)$$

In Equation (5) the subscript 1 corresponds to the position at the desired flow rate, and 0 corresponds to the zero point with no gas flow. It is this uncertainty that captures the impact of external noise on the accuracy of the thrust stand, as increased vibration leads to a larger standard deviation of the measured value at a given flow rate.

The uncertainty of the calibration is the combined uncertainties of the capacitance measurement, the capacitance curve fit, and the uncertainty of the voltage applied to the CFG plates.

$$U_{cal} = \sqrt{U_C^2 + U_{fit}^2 + U_V^2} \quad (6)$$

The uncertainty of the capacitance curve fit is estimated as

$$U_{fit} = 1 - R_{fit}^2$$

where R_{fit} is the coefficient of determination of the curve fit. The capacitance is measured using a BK Precision 889B LCR meter with an uncertainty of $\pm 2\%$.⁹ The voltage of the CFG plates is measured with an HP 974A multimeter with an uncertainty of 0.05%.¹⁰ The gas flow rate is measured using an Alicat MS series flow meter with an uncertainty of $\pm (0.4\%$ of reading + 0.2% full scale).¹¹ Substitution of the above equations into Equation (4) yields the total uncertainty of the thrust measurement. At low flow rates uncertainty is primarily driven by position uncertainty caused by noise. At higher rates, though noise is still a large factor, the system uncertainties from the calibration process are of a similar order of magnitude. For the data presented here, the average uncertainty is 14.2%, though this is heavily biased by the low signal-to-noise ratio observed at the lowest flow rate. A better representation is the median uncertainty of 5.05%. Further improvement of this uncertainty can be made by limited external vibration and increasing the accuracy of the capacitance measurement during calibration.

C.3 Vibration Isolation

Assuming that accurate and calibrated instruments are used, the largest contribution of uncertainty to the thrust measurement is variation of the thrust stand arm position due to mechanical noise. This is primarily generated by vibrations from the backing pumps transmitted through the manifold back to the chamber, although other external perturbations can contribute, such as heavy equipment operation, personnel passing nearby, etc. In order to provide high accuracy measurements, a system to reduce incident vibrations on the thrust stand is required. For simplicity and ease of design, a passive isolation system was chosen, which essentially behaves as a mass-spring-damper system. Such systems are characterized by the vibration transmissibility, T , which is the ratio of the amplitude of the acceleration of the test article to the incident acceleration on the system. For an ideal mass-spring-damper isolation system, the transmissibility as a function of vibration frequency, f , is given by the classical relation

$$T = \sqrt{\frac{1 + \left(2\zeta \frac{f}{f_n}\right)^2}{\left[1 - \left(\frac{f}{f_n}\right)^2\right]^2 + \left(2\zeta \frac{f}{f_n}\right)^2}} \quad (7)$$

where f_n is the natural frequency of the system and ζ is the damping ratio. These terms are defined as

$$\zeta = \frac{c_d}{4\pi m f_n} \quad (8)$$

$$f_n = \frac{1}{2\pi} \sqrt{\frac{k}{m}} \quad (9)$$

where c_d is the damping coefficient, m is the mass of the system, and k is the system stiffness. An example transmissibility curve is shown in Figure 4.

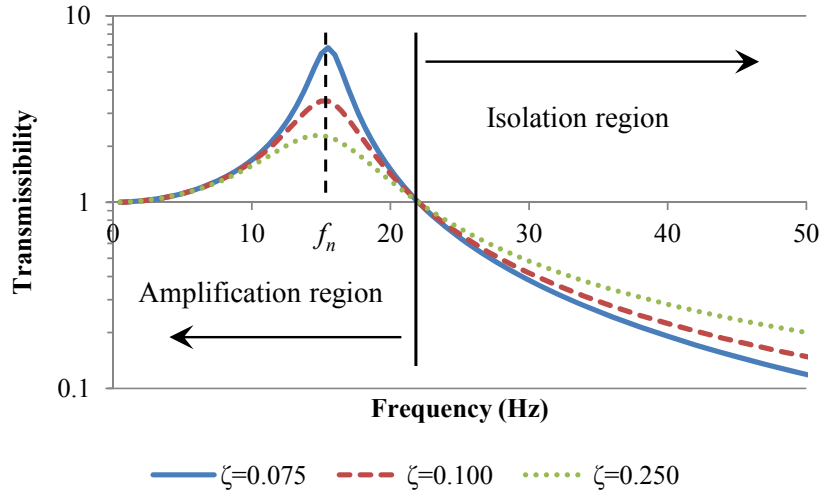


Figure 4. Conceptual transmissibility curve with $f_n = 15.5$ Hz.

While the behavior of this idealized isolation system diverges from empirical results at higher frequencies due to nonlinear effects, it does establish some key guidelines for designing vibration isolation systems. The first is that the behavior of the isolation system can be divided into two regions in the frequency spectrum: amplification and isolation regions. For frequencies below a threshold (generally $\sqrt{2}f_n$), transmissibility is greater than one and signifies that incident vibrations will be amplified, particularly those frequencies near the system natural frequency. For frequencies above this threshold, transmissibility is less than one and denotes a reduction of the amplitude of the oscillations at the test article compared to the incident vibrations. The second guideline is that the region of isolation can be increased by decreasing the system natural frequency. Since the mass of test articles is generally set by experimental requirements, the more flexible approach is to minimize system stiffness. This would take the form of softer elastomer mounts, a looser spring, or an elongated pendulum length, depending on the isolation system used. The third guideline is that increased damping tends to shift the transmissibility curve closer to unity. Thus for frequencies in the amplification region, increased damping reduces transmissibility; for frequencies in the isolation region, transmissibility is increased. Therefore, selection of total system damping is a balance between isolating the system from harmonic vibrations, which is usually covered by minimizing system stiffness such that the natural frequency is well below the fundamental frequency of the incident vibrations, and shocks, which tend to excite vibrations at the system natural frequency.

Using the guidelines established above, the vibration isolation system used consists of a spring suspension system inspired by a similar NIST vacuum vibration isolation system¹² with the addition of magnetic damping. The system consists of an aluminum optical table - to which the thrust stand is mounted - suspended by four extension springs with an individual stiffness of 1115 N/m. The extension springs are suspended from threaded rods which can be adjusted to control coarse leveling and height. The spring lengths are selected such that the natural frequency of the pendulum created by the

suspension system is small (~ 0.25 Hz) while maintaining a practical form factor. In order to provide sufficient damping for the system, neodymium magnets are mounted to the stationary support structure. Aluminum brackets are mounted on the optical table opposite the magnets to provide a surface where eddy current dissipation can occur. A picture of the thrust stand and vibration isolation system is shown in Figure 5.

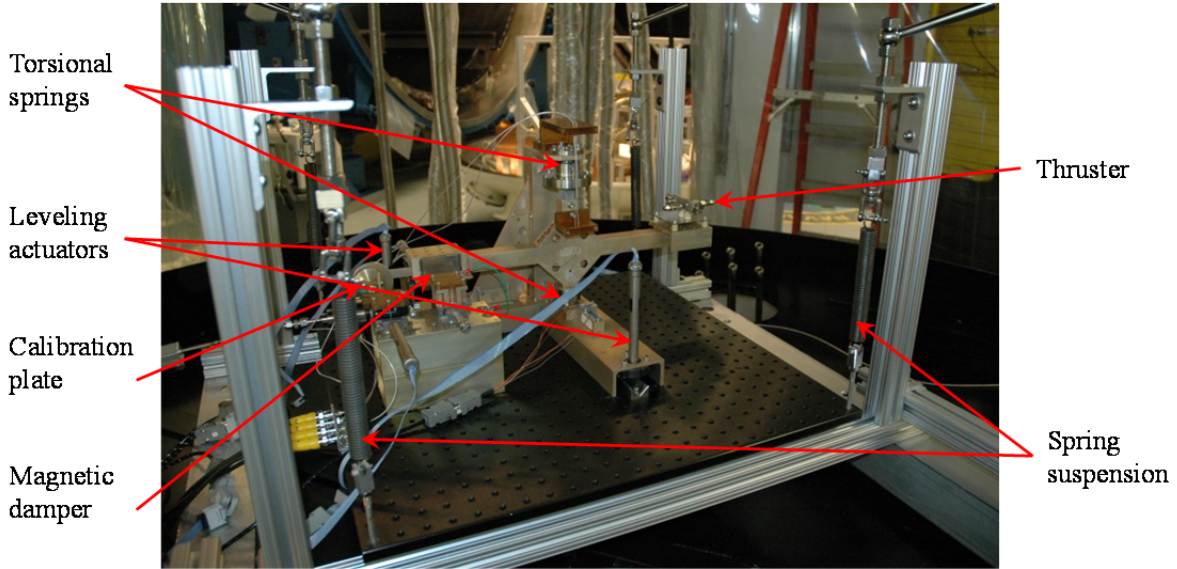


Figure 5. Torsional thrust balance with spring suspension vibration isolation.

To confirm the efficacy of the isolation system, four accelerometers are placed at various locations in the system to measure accelerations. One is located on the vacuum chamber surface, two on the optical table, and one on the thrust stand arm. Endevco model 2228C piezoelectric accelerometers are used in conjunction with an Oros OR35 multi-analyzer. The measured accelerations are used to calculate the transmissibility of the system shown below in Figure 6.

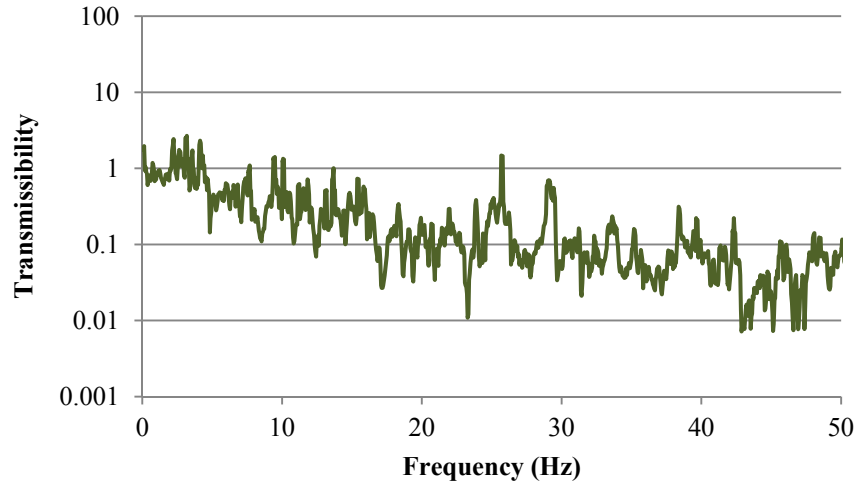


Figure 6. Thrust stand to chamber transmissibility.

III. Results

The three performance parameters of interest are the thrust, specific impulse, and efficiency of the nozzle. The specific impulse is given as

$$I_{sp} = \frac{F_T}{\dot{m}g} \quad (10)$$

where $\dot{m}$ is the mass flow rate of the propellant. The throat Reynolds number for a given flow rate is

$$Re^* = \frac{4\dot{m}}{\mu\pi d^*} \quad (11)$$

where μ is the viscosity of the propellant, and d^* is the throat diameter. From Equations (10) and (11) the thrust and specific impulse as a function of throat Reynolds number can be created, as shown in Figure 7. The nozzle performance and measured operating conditions are tabulated below in Table 1. Gas temperature is 293.96 ± 0.027 K.

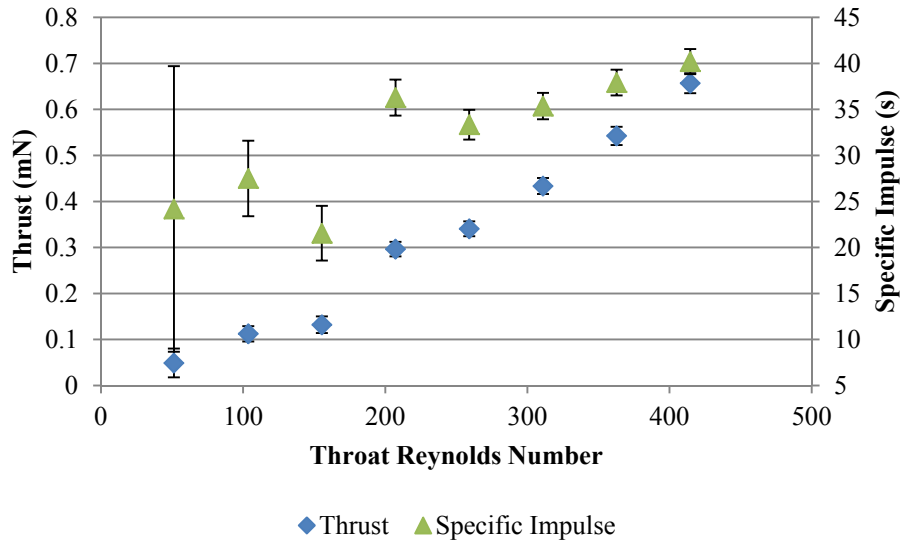


Figure 7. Thrust and specific impulse as a function of throat Reynolds number.

Table 1. Nozzle test bed operating conditions and performance.

Flow rate (sccm)	Re*	Chamber pressure (bar)	Thrust (mN)	I _{sp} (s)
9.9	51.3	0.0469	0.049 ± 0.031	24.2 ± 15.5
20.0	103.6	0.0571	0.112 ± 0.017	27.5 ± 4.1
30.0	155.4	0.0781	0.132 ± 0.018	21.5 ± 3.0
40.0	207.2	0.0976	0.296 ± 0.016	36.3 ± 2.0
50.0	259.1	0.1144	0.341 ± 0.016	33.3 ± 1.6
60.0	310.9	0.1310	0.434 ± 0.017	35.4 ± 1.4
70.0	362.8	0.1473	0.542 ± 0.020	37.9 ± 1.4
80.0	414.6	0.1625	0.657 ± 0.022	40.2 ± 1.4

As can be seen in Figure 7, there is higher uncertainty at very low Reynolds numbers. This is due to mechanical noise external to the vacuum chamber creating oscillations in the thrust stand arm, which for very low flow rates are of the same order as the arm displacement due to thrust. For higher flow rates this mechanical noise becomes comparatively smaller as thrust, and thus arm displacement, increases.

IV. Discussion

The efficiency of the nozzle is quantified as the ratio of the measured specific impulse to the theoretical specific impulse of an isentropic expansion through the same nozzle, also called the specific impulse efficiency.

$$\eta = \frac{I_{sp}}{I_{sp,s}}$$

The isentropic specific impulse is calculated from the isentropic nozzle relations, yielding

$$I_{sp,s} = \frac{1}{g} \sqrt{\frac{2\gamma}{\gamma-1} RT_0 \left[1 - \left(\frac{p_e}{p_0} \right)^{\frac{\gamma-1}{\gamma}} \right]} \quad (12)$$

where γ is the ratio of specific heats, R is the specific gas constant, p_0 and T_0 are the stagnation pressure and temperature of the propellant in the thrust chamber, respectively, and g is the acceleration due to gravity. Using Equation (12) and the specific impulse calculated previously, the nozzle efficiency is calculated and shown in Figure 8.

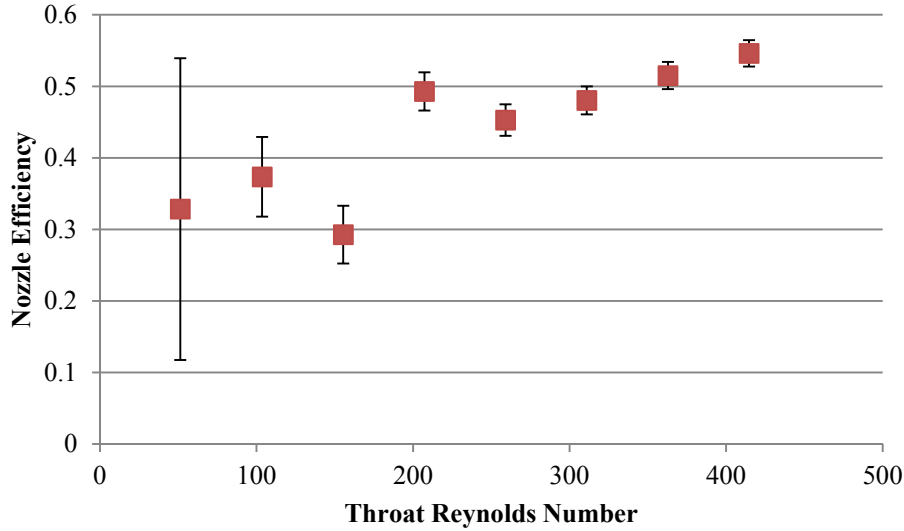


Figure 8. Nozzle efficiency as a function of throat Reynolds number.

Direct comparisons of small nozzle flow measurements with the literature are difficult due to the wide range of nozzle geometries used. The nozzle geometry could not be matched exactly to any previous studies, and, very few studies have examined throat Reynolds numbers below 500. There is general agreement within the literature about the qualitative behavior of the nozzle efficiency as a function of the throat Reynolds number: at Reynolds numbers above 3000, efficiency has relatively plateaued, and slowly decreases with decreasing Reynolds number until a marked drop in efficiency occurs at Reynolds numbers 500-1000.^{1,2,4,5} The exact nature of this sudden drop in efficiency is hard to discern as few sources report findings below a Reynolds number of 500, as accurate measurements of thrust are difficult past this point. As such, any comparison with pre-existing data is inexact. This is further compounded by the fact that the nozzle geometries differ between sources. The throat diameter in Whalen's nozzles is 0.06 inches, almost a factor of six higher than what is used here. This results in a higher mass flow rate through the nozzle in Whalen's experiment, and there may be some impact from delayed flow rarefaction. A closer examination must occur when performance data of the MRJ at higher Reynolds numbers has been taken that can be more directly compared to the literature. Future work will entail expanding the experimental dataset to cover the whole range of 100-3000 throat Reynolds number, as well as hot flows and other propellants.

V. Conclusions

In conclusion, a micro-resistojet has been tested on a torsional thrust balance system and its nozzle efficiency calculated for throat Reynolds numbers less than 500. The nozzle efficiency was found to be lower than expected compared to relatively similar nozzle data from a previous study, but direct comparison is inaccurate without matching throat Reynolds number flows. An expansion of the dataset will be conducted in future work to allow for better analysis of low Reynolds number phenomena.

References

- ¹Whalen, Margaret V., “Low Reynolds Number Nozzle Flow Study,” *NASA Technical Memorandum 100130*, 1987.
- ²Grisnik, Stanley P., Smith, Tamara A., and Saltz, Larry E., “Experimental Study of Low Reynolds Number Nozzles,” *NASA Technical Memorandum 89858*, AIAA-87-0992, 1987, pg 1-12.
- ³Ketsdever, Andrew., D., Calbough, Michael T., Gimelshein, Sergey F., and Alexeenko, Alina, “Experimental and Numerical Determination of Micropropulsion Device Efficiencies at Low Reynolds Numbers,” *AIAA Journal*, **43**, No. 3, 2005, pg 633-641.
- ⁴Murch, C. K., Broadwell, J. E. Silver, A. H., and Marcisz, T. J., “Low-Thrust Nozzle Performance,” *6th AIAA Aerospace Sciences Meeting*, New York, NY, January 1968, AIAA TP 68-91, pg 1-7.
- ⁵Bayt, Robert L. and Breuer, Kenneth S., “Viscous Effects in Supersonic MEMS-Fabricated Micronozzles,” *Proceedings of the 3rd ASME Microfluids Symposium*, Anaheim, CA, November 1998, pg 1-7.
- ⁶Osborn, M. F., Holman, T. D., Williams, L. T., Rosenberg, D. A., Tuttle, S., “Overcoming Low Nozzle Efficiency: A Test Correlated Numerical Investigation of Low Reynolds Number Micro-Nozzle Flow,” *51st AIAA Joint Propulsion Conference*, July 2015.
- ⁷Holman, T. D., Osborn, M. F., “Numerical Optimization of Nozzle Geometries for Low Reynolds Number Micro-Nozzle Flow,” *51st AIAA Joint Propulsion Conference*, July 2015.
- ⁸Rosenberg, D. A., Williams, B., Tuttle, S., Osborn, M. F., and Williams, L. T., “Optical Measurements of Density and Species Concentration in a Low Reynolds Number Micro-Nozzle Flow,” *53rd AIAA Aerospace Sciences Meeting*, January 2015.
- ⁹BK Precision, *Model 889B Bench LCR/ESR Meter Instruction Manual*, Manual #481-528-9-001B, 2009.
- ¹⁰Hewlett-Packard, *974A Multimeter User's Guide*, Manual #00974-90001, April 1994.
- ¹¹Alicat Scientific, *Operating Manual*, Revision 28, DOC-ALIMAN16, January 2014.
- ¹²Lawall, John and Kessler, Ernest, “Design and Evaluation of a Simple Ultralow Vibration Vacuum Environment,” *Review of Scientific Instruments*, **73**, No. 1, 2002, pg 209-215.